Simulation vs Practical Development: Technical Proposal

1. Recommendation (Summary)

I recommend a **hybrid approach**: use simulation for quick calculations and early design choices, build a cheap prototype within the first few days, and **trust measured results over the models**. Simulation alone delays the first flight and relies on assumptions we cannot check. Building alone wastes time and materials on problems a short calculation would have caught.

2. Assumptions

- Electric fixed-wing aircraft, span about 1.5 m, mass about 1.8 kg, 30 m runway, timed circuit; team of 4 to 6 students using hobby-grade parts.
- Air density 1.1 kg/m^3 (altitude and heat). Small wings fly at low Reynolds number (about 2×10^5), so published airfoil data are only approximate.

3. Simulation-Led vs Prototype-Led Design

Simulation-led. *Pros:* cheap and fast; stability and power can be checked before any wood is cut. *Cons:* results are only as good as the inputs. XFLR5 is weak at stall and prop wash, ANSYS CFD takes long to set up and run, and neither can see build errors or weak joints.

Prototype-led. *Pros:* tests the real aircraft, including build quality, and gives real data. *Cons:* every change costs days and materials, crashes destroy parts, and wind and pilot differences make causes hard to separate.

4. Comparison Table

Criterion	Simulation-led	Prototype-led	Hybrid (proposed)
Cost	Lowest; free tools (XFLR5, OpenVSP); ANSYS and SolidWorks need lab licences	High; rebuilds use foam, balsa, props	Low to moderate; 1 to 2 prototypes
Time	Fast per variant; CFD set-up is long	Slow per iteration; early first flight	First flight by day 10
Accuracy	Good for trends, rough on drag	Highest, but noisy	Model calibrated by data
Iteration speed	Minutes	Days	Minutes (geometry),

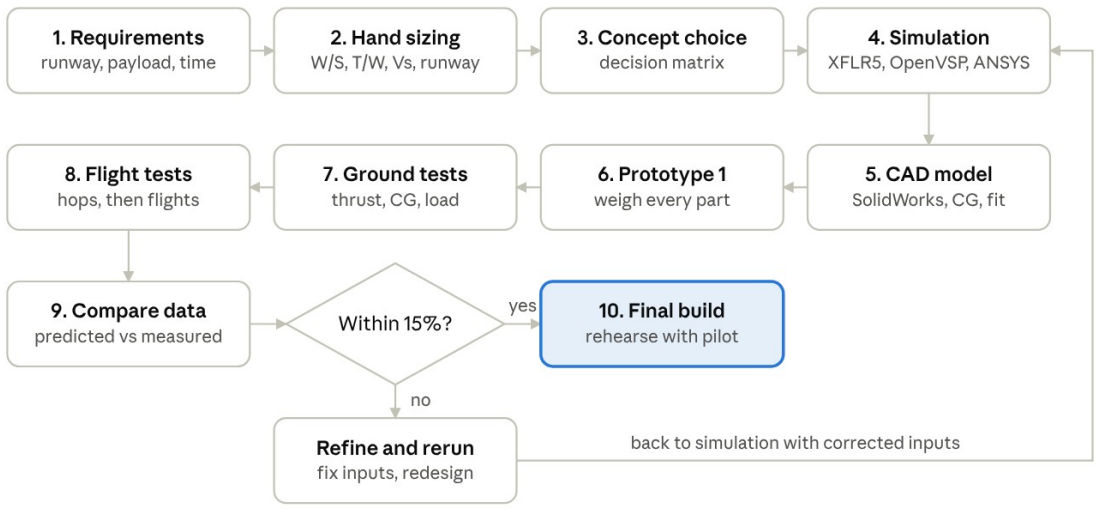
Criterion	Simulation-led	Prototype-led	Hybrid (proposed)
			hours (CG, prop)
Failure detection	Finds stability and power problems; misses build faults	Everything, but late and destructive	Simulation checks the design; ground tests check the build
Final performance	Tuned to the model, not reality	Reliable, but may not be the best design	Optimised on a validated model

5. Proposed Workflow

Steps 1 to 3 (paper): turn the runway, payload, time and scoring rules into numbers; hand-calculate wing loading, stall speed, thrust and takeoff distance; compare 2 to 3 layouts in a scoring table.

Steps 4 and 5 (computer): XFLR5 (wing and stability), OpenVSP (layout), eCalc (motor and power) and ANSYS (extra airflow checks); SolidWorks or Fusion 360 for the 3D model and cutting files, with SolidWorks Simulation or ANSYS for stress checks on the wing joiner and motor mount.

Steps 6 to 10 (build and test): a cheap Prototype 1 with every part weighed; ground tests (weight, CG, motor thrust, strength); flight tests (hops, then full flights, data recorded); compare with the model, fix its inputs and improve the design; then the final build and rehearsal with the real pilot and payload. The flowchart below shows the steps and the feedback loop.



development workflow · 10 steps, 1 decision, 1 feedback loop

6. Sizing Calculations (Worked Example)

- Weight 1.8 kg (17.7 N), wing area 0.36 m^2 (span 1.5 m), so wing loading = 5.0 kg/m^2.
- Stall speed $V_s = \sqrt{(2W / (\rho S C_{Lmax}))} \approx 8.3$ m/s, using $C_{Lmax} = 1.3$. Lift-off speed ≈ 1.2 $V_s \approx$ 9.9 m/s.
- Thrust needed ≈ 0.65 × weight: 12 N static, about 8 N average during the takeoff run.
- Ground roll: acceleration $a = (8 - 0.5 - 0.9) / 1.8 \approx 3.7$ m/s^2 (0.5 N drag, 0.9 N rolling friction), so distance = $V^2 / 2a \approx 13$ m (limit 30 m). If real thrust is only 60% of predicted, the distance rises to about 26 m, so thrust must be measured. (Method: Raymer; Anderson.)

7. Parameters: Simulate vs Verify Physically

Simulate	Verify physically	Why a physical test is needed
Stability (XFLR5)	Actual weight and CG	CAD ignores glue, wiring and covering
Airfoil and stall (XFoil), airflow (ANSYS)	Motor thrust and current	Real motors and batteries differ from datasheets
Power (eCalc), part stress (SolidWorks, ANSYS)	Strength, takeoff distance, control response	Models assume a perfect build

Our ANSYS Fluent run: a steady 3D simulation of the wing (SST k-ω model), used as a visual check on XFLR5 (Fig. 1), not for exact forces.

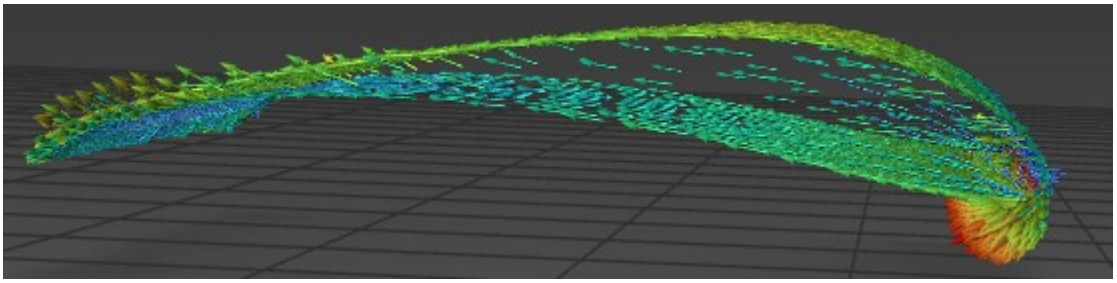


Fig. 1: Fluent velocity vectors on the wing surface, coloured by velocity magnitude.

8. Validating Simulations With Measurements

1. **Predict first**, with a tolerance (e.g. stall speed 8.3 m/s ± 10%).
2. **Measure** weight, CG, motor thrust, stall speed, takeoff distance and glide ratio; repeat each test at least three times.
3. **Compare and fix the inputs**, then rerun. Trust the model when it matches the data within about 10 to 15%.

9. Case: The Simulation Looks Good, but the Prototype Does Not Take Off

Plausible causes:

- *Mass and CG:* heavier than modelled, or CG too far forward or aft.
- *Propulsion:* thrust below 12 N from wrong prop, voltage sag, weak ESC or reversed prop.
- *Aerodynamics and ground:* low wing incidence, warps, CLmax below assumption, high rolling friction, too little elevator, tailwind or thin air.

Diagnostic plan (cheapest first, one change at a time):

1. Audit: weigh the aircraft; check CG, control directions and prop orientation.
2. Static thrust test at full throttle; below about 10 N, fix propulsion first.
3. Check wing and tail incidence with a digital level.
4. Recompute takeoff distance with measured mass and thrust; if it says no, the inputs were wrong, not the aircraft.
5. Taxi and hop tests, changing one variable at a time, then update the model.

10. Four-Week Schedule With Contingency

Week	Activities	Gate
1	Requirements, sizing, XFLR5/OpenVSP study, order parts, start Prototype 1	Design baseline frozen (day 5)
2	Finish prototype, ground tests, first flights by day 10 to 11, first calibration	Predicted vs measured table
3	Update model, build final airframe, second test cycle	Final airframe flown
4	Mission rehearsals, tuning, spares, documentation	Competition-ready aircraft

Contingency: about 20% (5 to 6 days): 2 days in week 2 for repairs or weather, 2 in week 3 for a rebuild, 1 to 2 in week 4. Long-lead parts are ordered in week 1 with spares; if a gate slips, cut scope, not test time.

11. References

1. Raymer, D. P., *Aircraft Design: A Conceptual Approach*, AIAA.
2. Anderson, J. D., *Aircraft Performance and Design*, McGraw-Hill.
3. Selig, M. S., *UIUC Low-Speed Airfoil Tests*; Drela, M., *XFOIL* (MIT); xflr5.tech; openvsp.org.

All numbers are assumed design estimates and must be replaced with competition rule values and measured data. AI tools were used for research and drafting; I reviewed all calculations and design choices.